ChatGPT

Proposed scheme: Independent-cumulant approximation

Carry at each layer: the full covariance **Σ**, the mean vector **μ**, and only the diagonal third cumulants **$d_3 \equiv \kappa_3(i,i,i)$**. (We ignore all off-diagonal third- and fourth-order terms.) We propagate these through the linear transform and ReLU nonlinearity as follows. First, propagate the linear layer:

- **Mean:** $\mu' = W\mu$.
- **Covariance:** $\Sigma' = W\Sigma W^T$. (This captures all second-order correlations.)
- **Diagonal third cumulant:** assuming inputs are independent beyond covariance, the third cumulant of a sum adds linearly【17†L161-L169】, so for each output neuron i

$$d'_{3,i} \;=\; \sum_j W_{ij}^3 \, d_{3,j} \,.$$

(Here we use $\kappa_3(Z_i+Z_j)=\kappa_3(Z_i)+\kappa_3(Z_j)$ under independence【17†L161-L169】.) We do **not** carry κ_4 or any off-diagonal κ_3 terms.

Next, to compute the post-ReLU means needed by the next layer, we treat each pre-activation as (approximately) Gaussian with its mean and variance. The expected ReLU output for Gaussian $X\sim N(\mu,\sigma^2)$ is given in closed form【23†L1199-L1202】:

$$E[\max(X,0)] \;=\; \sigma\,\phi\!\left(\tfrac{\mu}{\sigma}\right) \;+\; \mu\,\Phi\!\left(\tfrac{\mu}{\sigma}\right),$$

where φ,Φ are the standard normal PDF and CDF. We apply this per-neuron using μ_i and $\sigma_i^2 = (\Sigma')_{ii}$. (Higher cumulants like κ_4 would only enter via small correction terms that we omit.)

Cost: In n^3-unit terms, propagating Σ' costs one n^3 (matrix multiply) and propagating d_3 costs another n^3. Mean updates are $\ll n^3$. Total $\approx 2n^3$ products ($\ll 125u$), satisfying the budget.

Falsifier: A simple decisive check is to compare the scheme's predicted κ_3-channel against the teacher's data. For example, at layer 11 compute $Y_{21}(i,j)=\kappa_3(i,i,j)$. Our scheme assumes off-diagonal $\kappa_3=0$, so it predicts Y_{21} off-diagonals = 0. The teacher's Y_{21} at that layer is nonzero, so the relative Frobenius error on off-diagonals would greatly exceed the chain's ~5% tolerance. Thus a one-step test of Y_{21} at L11 (or similarly D_{21} at L15) would falsify this scheme by a wide margin. (In other words, neglecting cross-neuron third cumulants causes unacceptably large errors.)

Missing term: To approach the chain's accuracy one would need to restore the omitted cross-cumulant terms (the off-diagonal K31 entries). Including a low-rank or structured approximation of those $\kappa_3(i,i,j)$ contributions would close the gap.

References: We use cumulant additivity under independence【17†L161-L169】 and the Gaussian ReLU mean formula【23†L1199-L1202】.